

Module - II: Diffraction of X-rays

Introduction

X-rays are electro-magnetic wave with very short wavelength ($0.1 - 100 \text{ \AA}$). They have high penetrating power. X-rays are produced when fast electron incidents on matter.

Laue's Experiment

For diffraction of X-rays, the size of slit should be comparable to the wavelength of X-ray. Therefore, X-ray diffraction is not possible with grating surfaces. Von Laue suggested that X-ray diffraction can be observed by using crystals because crystals are regular arrangements of atoms with inter-atomic spacing of the order of 10^{-10} m .

Experimental Arrangement:

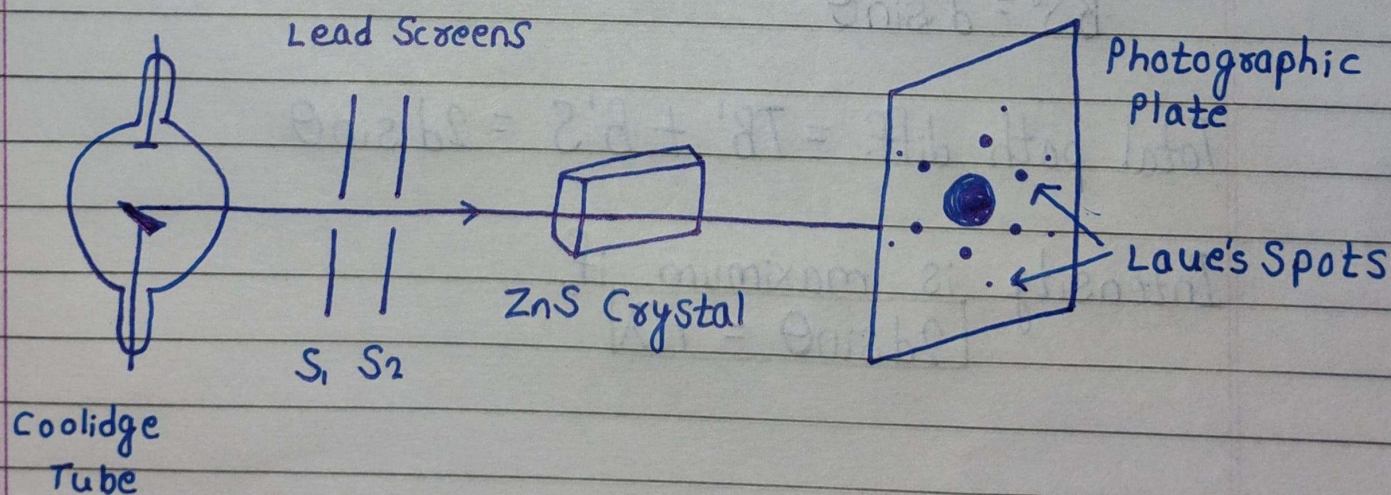
X-rays are produced by Coolidge tube. They are passed through fine holes into lead screens S_1 & S_2 . A fine beam of X-rays emerges out and incident on a zinc sulphide crystal. The diffracted beam is collected on a photo-graphic plate.

Observations:

Plate was exposed to X-ray for 40 minutes. After developing, it was found that there was a big bright centre spot. This spot is surrounded by a number of symmetrically arranged spots called Laue's spot. Each spot corresponds to a particular angle of diffraction.

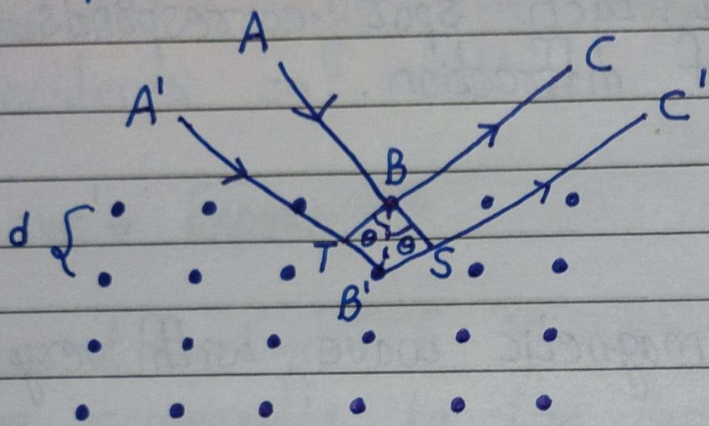
Imp. Outcomes:

1. X-rays are electro-magnetic wave with very short wavelength.
2. Diffraction of X-rays is possible by passing through crystal.
3. The atoms in a crystal are arranged in regular 3D array with inter-atomic distance of the order of $1 \text{ \AA}$.



Bragg's Law

When parallel monochromatic X-ray beam incident on a crystal it is reflected. According to Bragg's Law, intensity of reflected beam is maximum if path difference is $n\lambda$.



In $\triangle BB'T$,

$$\sin \theta = \frac{TB'}{BB'} = \frac{TB'}{d}$$

$$\Rightarrow TB' = d \sin \theta$$

Similarly, in $\triangle BB'S$,

$$B'S' = d \sin \theta$$

$$\text{Total path diff.} = TB' + B'S' = 2d \sin \theta$$

Intensity is maximum if,

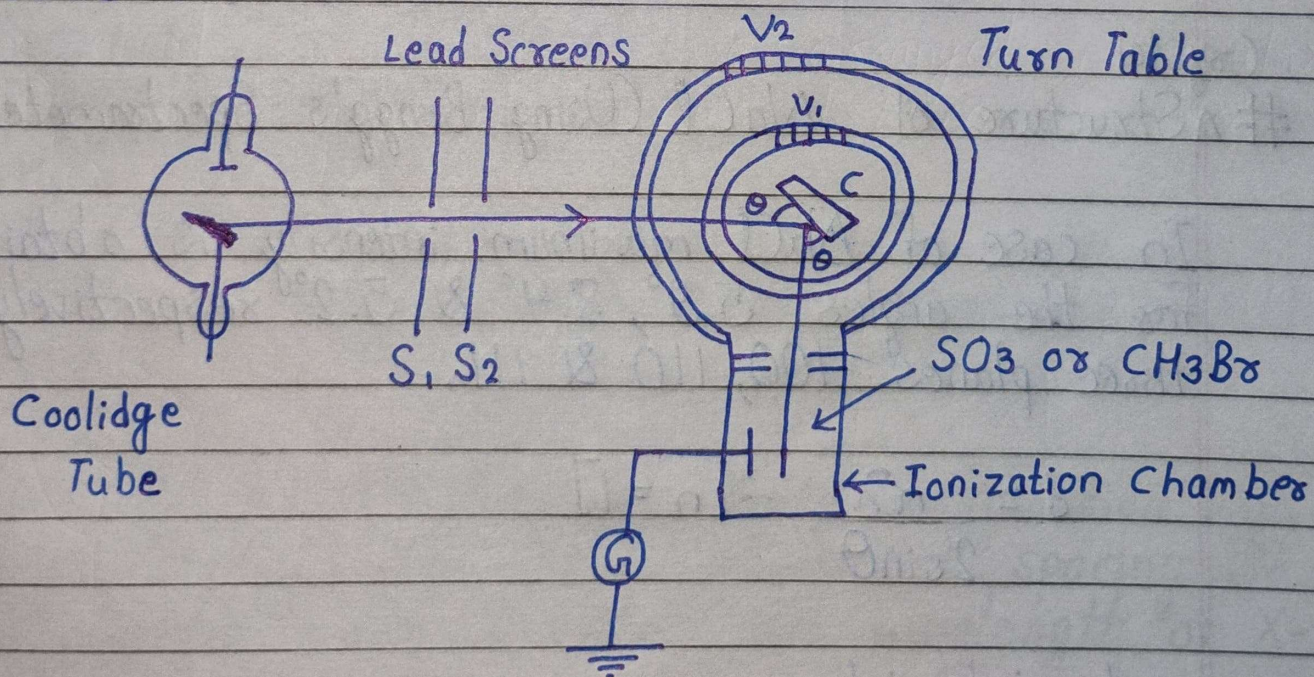
$$2d \sin \theta = n\lambda$$

Bragg's Spectrometer

Experimental Arrangement:

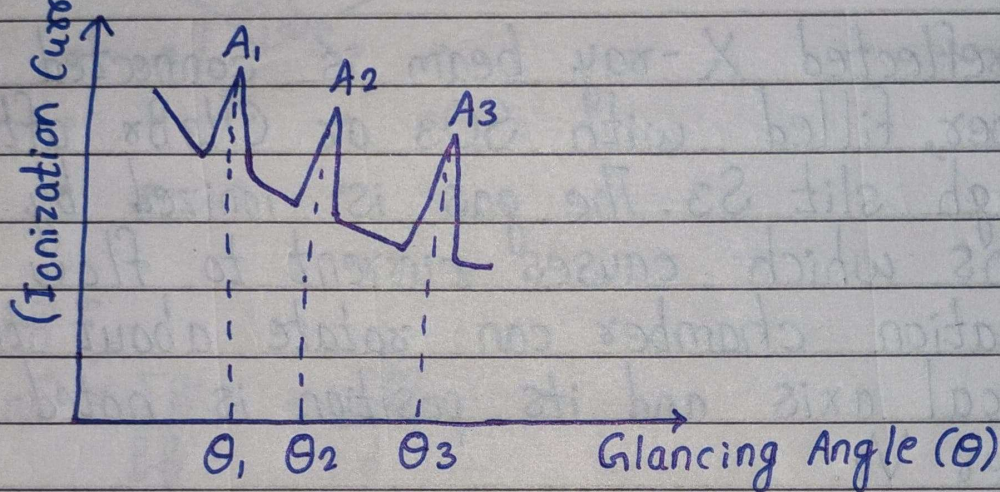
The crystal 'C' is mounted on a prism table which can rotate about a vertical axis. Its position can be noted by a vernier V_1 . Beam of X-ray is incident on crystal after passing through lead screens S_1 & S_2 .

The reflected X-ray beam is connected in ionization chamber, filled with SO_3 or CH_3Br after passing through slit S_3 . The gas is ionized by X-ray photons which causes current to flow in galvanometer. Ionization chamber can rotate about the same vertical axis and its position is noted by another vernier V_2 .



Observation:

The ionization chamber is adjusted to get maximum intensity. A graph is plotted between glancing angle and ionization current. The values of current increases abruptly for certain values of θ . The angles θ_1, θ_2 & θ_3 are obtained from the graph. The ratio $\sin \theta_1 : \sin \theta_2 : \sin \theta_3 = 1 : 2 : 3$, which is in consistent with Bragg's law.



Crystal

Structure of NaCl (Using Bragg's Spectrometer)

In case of NaCl, maximum intensity is obtained for the angles $5.9^\circ, 8.4^\circ$ & 5.2° respectively using three planes 100, 110 & 111.

$$d = \frac{n\lambda}{2\sin\theta} \quad [n=1]$$

$$\begin{aligned} & d_{100} : d_{110} : d_{111} \\ &= \frac{1}{\sin 5.9} : \frac{1}{\sin 8.4} : \frac{1}{\sin 5.2} \end{aligned}$$

$$= \frac{1}{0.103} : \frac{1}{0.146} : \frac{1}{0.09}$$

$$= 1 : 0.705 : 1.14$$

$$= 1 : \frac{1}{\sqrt{2}} : \frac{2}{\sqrt{3}}$$

$\Rightarrow$ NaCl is fcc lattice.

Numericals

Q. Calculate the longest wavelength that can be analysed by a rock-salt crystal of spacing $2.82 \text{ \AA}$ in second order?

Solⁿ:

$$n = 2$$

$$d = 2.82 \text{ \AA}$$

$$2d \sin \theta = n \lambda$$

$$\Rightarrow \lambda_{\max} = \frac{2d}{n} = 2.82 \text{ \AA} \quad [\text{For maximum } \sin \theta = 1]$$

Q. In Bragg's reflection of X-ray, a reflection was found at 30° with lattice planes of spacing $1.87 \text{ \AA}$ in second order. Calculate the wavelength of X-ray?

Solⁿ:

$$\theta = 30^\circ, d = 1.87 \text{ \AA} \text{ \& } n = 2$$

$$\therefore \lambda = \frac{2d \sin \theta}{n} = 0.935 \text{ \AA}$$

9) The first order reflection from the plane of NaCl is obtained at $2\theta = 20^\circ$. If 'd' is $2.82 \text{ \AA}$, calculate the wavelength of X-ray used?

Solⁿ:

$$\theta = 10^\circ, d = 2.82 \text{ \AA} \text{ \& } n = 1$$

$$\therefore \lambda = \frac{2d \sin \theta}{n} = 0.9793 \text{ \AA}$$